A-level

ENVIRONMENTAL SCIENCE

Questions by topic: Pesticides - Set 1

Mark scheme

AQA spec 3.4.3.2.8; Pollution; textbook pp. 262-266

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**Calculation** 1 mark for the correct answer; working does not have to be shown. **1** 25 g ha^{-1} (accept 24-27) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (g ha^{-1}) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	1
01.2	1 mark for the point below. **1** The highest dose that can be applied without killing any members of the population / any individuals of the (non-target) species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
01.3	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** The MDAF (minimum dose always fatal) of the pest is about 50 g ha^{-1} **1** This is higher than the MDNF of the bee (about 20 g ha^{-1}) **1** So a dose that kills all pests kills some bees / the two curves overlap between about 20 and 50 g ha^{-1} **1** A dose of 30-40 g ha^{-1} would kill most pests with limited bee deaths (compromise) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
01.4	**Any 2 of 5** 1 mark each for any 2 of the 5 points below, in any order, maximum 2. **1** Spray at night / when bees are not flying **1** Spray when the crop is not in flower **1** Spray on still days to reduce spray drift **1** Use a systemic pesticide / seed coating rather than spraying **1** Use smaller droplets for more even coverage **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
01.5	**Any 3 of 5** 1 mark each for any 3 of the 5 points below, in any order, maximum 3. **1** Specificity is the extent to which the pesticide is toxic only to the target species / range of species affected **1** A highly specific pesticide has low toxicity to non-target species (eg bees, fish, mammals) **1** So fewer non-target deaths / less loss of predators, pollinators **1** Fewer indirect (food chain) effects / fewer inter-species relationships lost **1 Example:** pyrethroids have high insect toxicity but low mammal toxicity **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-263; AQA spec 3.4.3.2.8.	

Do not write outside the box

Question	Marking guidance	Marks
02.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** High persistence means DDT remains in the environment for years / decades, so it has time to disperse and be taken up **1** High liposolubility means it dissolves in fat and is stored in body tissues rather than excreted (bioaccumulation) **1** Concentration increases at each trophic level as predators eat many contaminated prey (biomagnification) **1** Birds of prey are at the top of the food chain so receive the highest concentrations **1** Sub-lethal doses caused thin eggshells that broke / reduced breeding success – higher doses caused sterility or death **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **bioaccumulation** → a pollutant building up inside one organism over time • **trophic level** → the feeding level in a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
02.2	**1 mark each** 1 mark for each of the 2 points below. **1 Advantage:** low persistence / low liposolubility so they do not bioaccumulate or biomagnify / break down quickly **1 Disadvantage:** high mammalian toxicity so farm workers / humans are at risk of acute poisoning (neurotoxin inhibiting acetylcholinesterase) – chronic effects (memory loss, depression) – suspected carcinogens **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	2
02.3	1 mark for the point below. **1** Pyrethroids are highly toxic to fish (even though mammal toxicity is low) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
02.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Neonicotinoids are water soluble so they are absorbed by the roots (from the soil) and translocated through the plant (systemic) **1** They are moderately persistent so remain in the soil long enough to be absorbed **1** Seed coating avoids spraying so no spray drift / no dust onto flowers or surrounding habitats **1** So bees / pollinators are less exposed than when the crop is sprayed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-265; AQA spec 3.4.3.2.8.	
03.1	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** 38.0 / 0.002 **1** = 19 000 (19 000 times / 1.9×10^4) **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (times) is already printed on the answer line. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks. • A bare wrong answer with no working. • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
03.2	1 mark for the point below. **1** Biomagnification (accept bioaccumulation as part of the process only if biomagnification is also implied by increase up the food chain) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biomagnification** → concentration rising at each step along a food chain • **bioaccumulation** → a pollutant building up inside one organism over time **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	1

Do not write outside the box

Question	Marking guidance	Marks
03.3	**1 mark each** 1 mark for each of the 4 points below. **1 Persistence:** the pesticide is not broken down (by light, microbes, enzymes) so it remains in the environment / organisms long enough to pass along the food chain **1 Liposolubility:** the pesticide dissolves in lipids / fat so it is stored in body tissue and not excreted in (water-based) urine **1** Each organism eats many organisms from the level below and retains the pesticide so concentration rises at each trophic level **1** Herons have long lives so accumulate over time **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Persistence** → does not break down / stays in the environment for a long time • **trophic level** → the feeding level in a food chain **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
03.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Organochlorines killed non-target insects (bees, butterflies) and top predators (birds of prey, otters) through biomagnification **1** They persist for decades so the damage continued long after use **1** Use in houses (on walls / nets) does not enter food chains / no runoff into water, so does not biomagnify **1** Benefit of preventing malaria deaths outweighs the risk (risk:benefit) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-264, 266; AQA spec 3.4.3.2.8.	

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Question	Marking guidance	Marks
04.1	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • Neonicotinoids are neurotoxins similar to nicotine – they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • **Very toxic to bees:** sub-lethal doses reduce navigation ability, foraging and immunity to disease – linked to colony decline • **Water soluble and moderately persistent:** remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil – broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • **Control:** seed coating rather than spraying – spraying when crops are not in flower / at night – still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops) – use of alternative pest control (cultural, biological, integrated control) • **Evaluation:** reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators – trade-off with crop yields **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content. • **persistent** → does not break down / stays in the environment for a long time • **Synergism** → two pollutants together doing more damage than the sum of each alone **Do not credit**	9
	• pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

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Question	Marking guidance	Marks
	Neonicotinoids are neurotoxins related to nicotine that block the neurotransmitter acetylcholine. They are highly toxic to insects but of low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier - the specificity that made them popular. Bees, however, are insects: even sub-lethal doses impair navigation, foraging and immunity, and this has been linked to colony decline. Because the compounds are water-soluble and moderately persistent they remain in the soil, are absorbed by roots and translocated through the plant (systemic action), so they appear in pollen and nectar that bees collect; persistence may also allow build-up in soils and aquifers, although sunlight and soil microbes break them down. Some fungicides act synergistically with them, increasing toxicity to bees.	
	Exposure is greatest when crops are sprayed or when dust from coated seeds is blown by wind, so impacts are reduced by using seed coatings instead of sprays, spraying only when crops are not in flower or at night when bees are not foraging, and applying on still days to limit drift. The EU has banned outdoor use on flowering crops, and integrated control with cultural and biological methods reduces the need for insecticides altogether.	
	These measures reduce bee exposure, but the alternatives farmers turn to, such as pyrethroids, also harm pollinators, and restrictions can reduce yields, so the trade-off between crop protection and pollination services remains.	
	Weaker answer - Level 2 (4 marks)	
	Neonicotinoids are insecticides that kill insects by affecting their nervous system. They are systemic so they get into the whole plant including the pollen, and bees pick them up when they collect pollen and nectar. This harms bees and can cause colonies to die. To reduce the impact, farmers can spray when crops are not flowering, and some neonicotinoids have been banned in the EU. Other methods of pest control could be used instead.	
	What makes the difference	
	• Level 2 links the systemic property to bee exposure correctly but omits the other properties (solubility, persistence, specificity via the blood-brain barrier, synergism) that the question's 'properties' demands.	
	• Level 3 explains the sub-lethal effects (navigation, foraging, immunity) rather than 'harms bees'.	
	• Level 3 includes the trade-off evaluation (alternatives also harm pollinators, yield costs) that the mark scheme rewards.	

Do not write outside the box

Question	Marking guidance	Marks
04.2	**Any 1 of 6** 1 mark each for any 1 of the 6 points below, in any order, maximum 1. **1** Biological control (introduced predators / pathogens) **1** Predator habitats (hedgerows, beetle banks) **1** Crop rotation **1** Companion / barrier crops **1** Sterile male technique / pheromone traps **1** Genetic resistance **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **Biological control** → using a natural predator, parasite or pathogen of the pest **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	
05.1	1 mark for the point below. **1** The dose that kills 50% of the (test) population **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
05.2	**1 mark each** 1 mark for each of the 3 points below. **1 Insecticide X:** LD50 for the beetle (6) is very similar to that of the pest (5) so it has low specificity / kills the predator as well **1** Loss of the predator removes natural control so the pest population may recover faster / increase (resurgence) **1** Y has a much higher LD50 for the beetle (150) so few predators killed / more specific **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **LD50** → the dose that kills half of a test population **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	3

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Question	Marking guidance	Marks
05.3	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** High water solubility so it is leached into rivers / groundwater / runoff **1** Toxic to aquatic organisms downstream **1** Mobile so it can affect areas away from the field **1** Washed off the crop quickly so more frequent application needed **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **leached** → nutrients or ions washed down out of the soil / washed away by water **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
05.4	**1 mark each** 1 mark for each of the 2 points below. **1** Long half-life = high persistence so it is not broken down quickly **1** So it remains in the organism / environment long enough to build up in tissues (especially if liposoluble) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **half-life** → the time for the activity, or the amount, to fall by half • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • 'after two half-lives nothing is left' - each half-life halves what remains, so some activity always continues	2
05.5	**Any 2 of 4** 1 mark each for any 2 of the 4 points below, in any order, maximum 2. **1** Absorbed and translocated within the plant so protects new growth **1** Not washed off by rain **1** Does not need to cover all leaf surfaces / less spray needed **1** Less spray drift so less harm to non-target species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first two - the question asks for two, so a list of extras cannot pick up the marks.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-264; AQA spec 3.4.3.2.8.	
06.1	1 mark for the point below. **1** The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time	1

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Question	Marking guidance	Marks
06.2	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1 Light:** UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions **1 Temperature:** higher temperatures increase the rate of chemical breakdown / microbial decomposition **1 Soil microbes / oxygen:** aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils **1** pH / presence of other chemicals may increase or decrease the rate of degradation **1** Adsorption onto clay / organic matter may protect from degradation and increase persistence **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistence** → does not break down / stays in the environment for a long time • **Adsorption** → nutrient ions sticking to the surfaces of clay and humus **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
07.1	**1 mark each** 1 mark for each of the 2 points below. **1** Numbers fell from about 210 pairs in 1945 to a minimum of about 25 in 1965 (steepest fall 1955-1960) **1** Numbers then recovered / rose steadily after 1970, reaching about 190 by 1985 (not yet back to 1945 level) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
07.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1** (200 - 25) / 200 x 100 **1** = 87.5% **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (%) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2

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Question	Marking guidance	Marks
07.3	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** DDT is persistent (remains for decades) and liposoluble so it bioaccumulates in prey (small birds) and biomagnifies in sparrowhawks (top predator) **1** Sub-lethal doses caused thin eggshells / reduced breeding success so the population declined even without adult deaths **1** Higher doses caused sterility / death **1** After restrictions, environmental concentrations slowly fell (persistence explains the delay before recovery) **1** Breeding success recovered so the population rose **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **persistent** → does not break down / stays in the environment for a long time	4
07.4	**Any 2 of 3** 1 mark each for any 2 of the 3 points below, in any order, maximum 2. **1** DDT persists in soil / sediments / food chains for many years so exposure continued **1 Birds of prey are k-selected:** few young per year, slow to mature so populations grow slowly **1** Other pressures (habitat loss, persecution) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	
08.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B - Low liposolubility and low persistence **Also accept** • **persistence** → does not break down / stays in the environment for a long time **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	1

Do not write outside the box

Question	Marking guidance	Marks
08.2	**1 mark each** 1 mark for each of the 2 points below. **1 Bioaccumulation:** build-up of a pollutant in the tissues of an individual organism over time because uptake is faster than excretion / breakdown **1 Biomagnification:** increase in concentration of the pollutant at successive trophic levels of a food chain **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **bioaccumulation** → a pollutant building up inside one organism over time • **biomagnification** → concentration rising at each step along a food chain **Do not credit** • answers that swap the two terms - bioaccumulation is the build-up inside one organism, biomagnification is the rise along a food chain • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** They are intended to be toxic (to the target species) **1** They are deliberately released / spread into the environment (rather than being an accidental or waste release) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 262-266; AQA spec 3.4.3.2.8.	

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